Crop Yield Estimation at Field Scales by Assimilating Time Series of Sentinel-2 Data Into a Modified CASA-WOFOST Coupled Model

Fujiang Ji, Jihua Meng, Zhiqiang Cheng, Huiting Fang, and Yanan Wang

***Abstract*— Rapidly and accurately estimating yields at field scales are very significant. Each type of currently yield estimation model has been well studied, yet all of them have certain limitations. Based on a coupled Carnegie-Ames-Stanford approach (CASA)-World Food Studies (WOFOST) model and time series Sentinel-2 imagery, we achieved daily crop simulations and crop yield estimations at two adjacent farms in China. The results indicated that the coupled model inherited the high computing speed of light use efficiency (LUE) models and the mechanistic advantages of crop growth models. The R^2 of yield simulation was 0.64 for the CASA model, 0.84 for the coupled model, and 0.86 for the WOFOST model, and the root mean square error (RMSE) values were 948.32, 792.11, and 623.64 kg/ha, respectively. The operating times of the CASA model, CASA-WOFOST coupled model, and WOFOST model over the growing season of wheat were 37 min, 48 min, and 1 day 5 h 7 min, respectively. Compared with the WOFOST model, the coupled model provided a much faster running speed in yield simulations and a similar accuracy; therefore, the proposed model can be applied for assessments at large farms with high-spatial-resolution images to obtain accurate yield simulations. Compared with the CASA model, the coupled model provided higher simulation accuracy in mountainous areas and regions of uneven terrain. It can be concluded that the proposed coupled CASA-WOFOST model can improve the precision, reliability, and stability of crop yield estimation; provide theoretical support for crop yield estimation at field scales; and promote the development of precision agriculture.**

***Index Terms*— Carnegie-Ames-Stanford approach (CASA) model, data assimilation, high temporal resolution normalized differential vegetation index (NDVI), remote sensing (RS), world food studies (WOFOST) model, yield estimation.**

I. Introduction

ACHIEVING a satisfactory crop yield is the ultimate goal of farming; the yield is the most direct indicator of the productivity of farmland and benefits to farmers. In addition, rapid and accurate yield estimation at the field scale is not only helpful for field management but also important for evaluating the response of farmland ecosystems to climate change, establishing scientific and rational food policies, optimizing external food trade, and so on [1]–[3].

After decades of developing crop yield estimation models, the current models can be divided into four categories: empirical statistical models, light use efficiency (LUE) models, crop growth models, and coupled models. First, empirical statistical models, such as agrometeorological models [4]–[8] and vegetation index (VI)-remote sensing (RS) statistical models [9]–[11], are simple; additionally, the required data are easy to obtain, thus directly linking the yield with impact factors with high regional accuracy. However, this type of model is generally simple and lacks a physical mechanism. Moreover, as the number of relevant impact factors increases, it can be difficult to find a stable statistical approach [12]. Second, LUE models are simple and practical. This type of model, which uses remotely sensed data to simulate the aboveground biomass, relies on a harvest index (HI) to estimate crop yields. LUE models are suitable for large-scale applications, and some representative models include Carnegie-Ames-Stanford approach (CASA), Cflux, eddy covariance-LUE (EC-LUE), global production efficiency model (GLO-PEM), vegetation photosynthesis model (VPM), vegetation photosynthesis and respiration model (VPRM), and so on. [13]–[16]. Some researchers have achieved high accuracy in regional winter wheat crop estimation through improved CASA models [17]–[19]. Crop growth models simulate the entire process of crop growth and the corresponding relationship with the surrounding environment from the perspective of ecological physiology, thereby integrating crop, soil, climate, management, and other factors to fully simulate physiological

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processes, such as crop respiration, photosynthesis, phenology, biomass accumulation, and the crop distribution, and estimate crop yields. The mainstream crop growth models include the AquaCrop model [20]–[22], ARID CROP model [23], World Food Studies (WOFOST) model [24]–[26], agricultural production systems simulator (APSIM) model [27]–[29], decision support system for agrotechnology transfer (DSSAT), HERMES, and the copping systems simulator (CROPSYST) model [30]–[32]. Different crop models have different driving factors. For example, the AquaCrop and soil water atmosphere plant (SWAP) models are soil-water-driven models, which emphasize the impact of water stress on crop growth and yield, and can simulate the impact of crop irrigation on crop yield. The APSIM, CROPSYST, and HERMES models are process-based models that take a lot more than soil moisture and salinity (N, P, etc.) into account, which contain a diverse range of soil processes such as water balance, nitrogen, and phosphorus transformation [28]. The WOFOST and simple and universal crop simulator (SUROS) models are driven by CO_2 and focus on the effects of CO_2, water, and temperature on yield. A coupled model combines two or more models with different principles and types to potentially avoid the shortcomings of the original models and take advantage of the benefits of each model to improve the accuracy of simulations and stability of the modeling system as well as reduce model operation costs. Some common coupling methods are as follows: 1) assimilating time series of remotely sensed data into crop growth models [33]–[38]; 2) coupling radiative transfer and crop growth models [39]–[41]; 3) coupling hydrological models and crop growth models [42]–[47]; and 4) combining crop growth models [48]–[50].

The current primary estimation models, especially LUE and crop growth models, have been extensively studied, yet they still have certain limitations. For example, the CASA model is relatively simple and practical, and it requires few parameters to set and calibrate. However, we can just simulate the net primary productivity (NPP) with the CASA model, to estimate the final crop yield, we need to construct a HI to achieve the conversion between the NPP and crop yield, but from photosynthetic assimilation to biomass accumulation and the final crop yields involve many complex physiological and biochemical processes that cannot be encompassed by a HI through a simple formula. The WOFOST model can effectively simulate the various stages of crop growth and development and be applied for field management. The WOFOST model simulates crop assimilation and respiration to estimate biomass accumulation and then distributes the biomass to stems, leaves, roots, and storage organs through partitioning coefficients. Therefore, this type of model has a strong mechanistic advantage in simulating crop yields with high accuracy. However, many parameters are required to perform detailed simulations, some data are expensive to obtain, and parameters can be difficult to calibrate.

In theory, by inputting the biomass simulated by a LUE model into a crop growth model and deducting the crop respiratory consumption, the storage organ partitioning coefficient can be used to obtain the final crop yield; this approach can replace the nonmechanical HI. A coupled model provides a higher simulating accuracy than that of the LUE model and utilizes the mechanism of a crop growth model. Additionally, a coupled model can effectively use remotely sensed data and the advantages of the two original models. In this article, taking the Sheltara and Tenihe Farms in Hulunbuir, Inner Mongolia Autonomous Region, as the research areas, based on time series of remotely sensed data, we adopt the crop growth simulation module in the WOFOST model to construct the CASA-WOFOST coupled model by replacing the HI in the CASA model. Then, we assess the accuracy and stability of the new yield estimation model and estimate the model running cost.

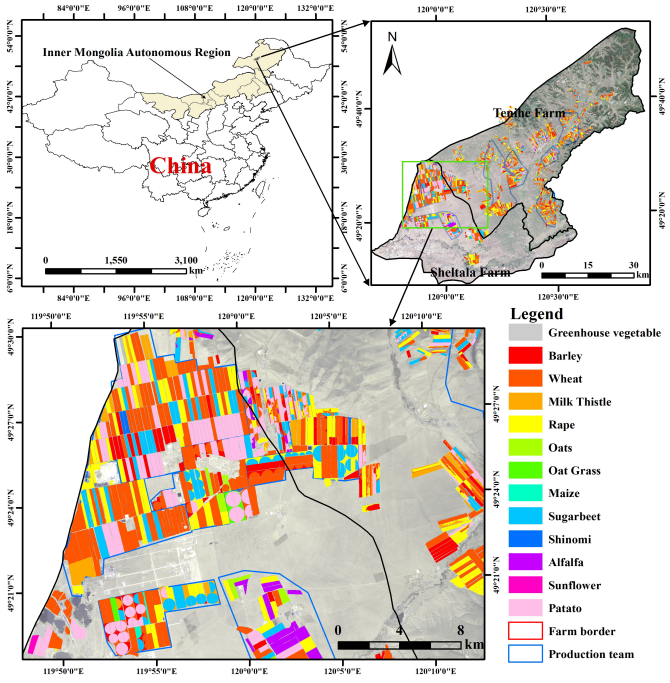

Fig. 1. Locations of the Sheltala and Tenihe Farms and the crop type distribution map. The background is the Sentinel-2A imagery acquired on June 19, 2019, shown in a quasi-natural color band composition of R(4) G(3) B(2) with standard deviation stretching.

II. Materials and Methods

A. Study Sites

This study selected Sheltala Farm and Tenihe Farm as the study area. The farms are located in Hulunbuir, northeastern Inner Mongolia. Sheltala Farm is located at 119° 65' −119 °60' E and 49 °06' −49° 30' N, and Tenihe Farm is located at 119° 45' −120 °55' E and 49 °13' −49° 50' N, with an average altitude of 650 m. Additionally, Sheltala Farm has an area of 12 000 ha served by water-saving irrigation, making it the largest water-saving irrigation farm in the Hulunbuir Reclamation Area. In addition, the type of soil in the farm area is black soil with a depth of 0.5–1.5 m, and the field area at the farm is large, making it suitable for RS technology applications at the field scale. Wheat, rape, milk thistle, and potato are the main crops in the study area. Fig. 1. shows the distribution of major crops at the Sheltala and Tenihe Farms in 2019.

B. Data Sources and Processing

1) Field Campaign: The field campaigns were conducted in April, June, July, and August 2019, which cover the entire

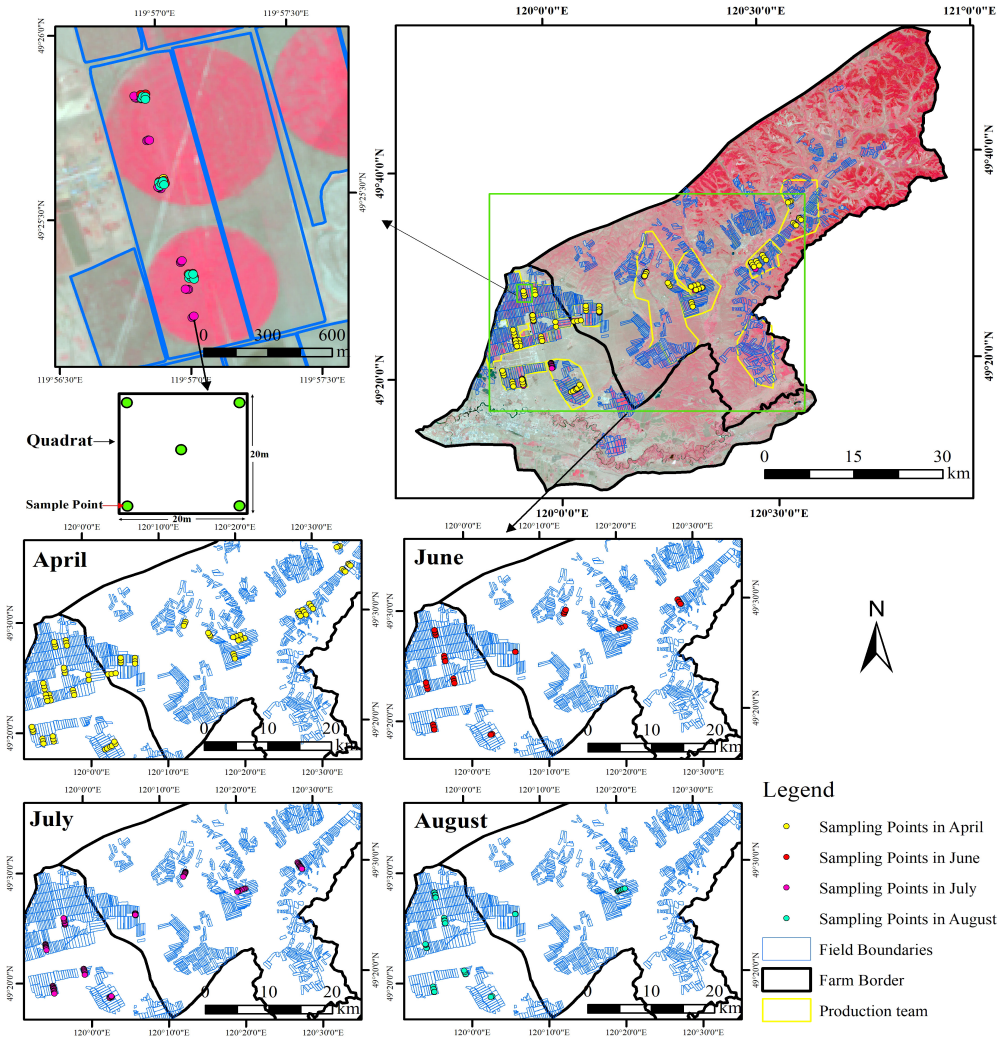

Fig. 2. Distribution of sampling points during the growing season. The background is the Sentinel-2A imagery acquired on June 19, 2019, shown in a false color band composition of R (8) G (4) B (3), with standard deviation stretching.

TABLE I
OBSERVATION DATA AND QUADRATS DURING THE GROWING SEASON

Field campaign date	Soil samples	VWC	LAI	Biomass	Yield
21-26 April 2019	80	80	0	0	0
16-20 June 2019	27	108	24	0	0
21-24 July 2019	0	0	16	59	0
16-20 August 2019	22	0	0	0	17

growing season of spring wheat. During the field campaigns, the soil samples are collected, and then the pH, available nitrogen, available phosphorus, and available potassium content in the soil samples were obtained from laboratory tests. Also, volumetric water content (VWC), leaf area index (LAI), crop growth period, biomass, and yield were observed. The campaign locations and observation data are shown in Fig. 2. and Table I, respectively. The observed data were mainly used for model calibration and accuracy evaluation. For example, the soil available nutrient, LAI, VWC, and fertilization data were used for model calibration; LAI was also used to evaluate the accuracy of LAI retrieval, and VWC was used to test the accuracy of VWC retrieval.

Methods applied during these field campaigns are shown in Fig. 2. Rectangular quadrats of 20 × 20 m were established with five sampling points in each (four points were located at the four corners of the quadrat, and the last point was located at the center of the quadrat). The mean value of the points in a quadrat was calculated as the result for that quadrat. The soil samples were collected from 0 to 20 cm depth by using a 20-cm-long soil extractor. Then, the soil samples from the five sampling points in the quadrat were fully mixed as the soil sample of that quadrat. There were two methods for VWC measurement. First, TDR300 equipment was used to collect VWC data at a depth of 0–20 cm in the soil. Additionally, two sets of VWC monitoring sensors were deployed at Sheltala Farm and Tenihe Farm, and each set included three probes, which were buried at 20, 40, and 60 cm below the ground and collected data every half hour. The LAI of the wheat canopy was determined with an LAI-2200. Biomass collection involved obtaining crops through a 0.25 m × 0.25 m square wire loop inside the quadrat. All the crops in the wire loop

TABLE II
SPECTRAL BAND CHARACTERISTICS OF SENTINEL-2 SATELLITE IMAGERY [52]

Band	Central Wavelength (nm)	Spectral width(nm)	Spatial Resolution (m)	Purpose
B1-Coastal	443	20	60	Atmospheric correction (aerosol scattering).
B2-Bule	490	65	10	Sensitive to vegetation senescing, carotenoid, browning and soil background; atmospheric correction (aerosol scattering).
B3-Green	560	35	10	Green peak, sensitive to total chlorophyll in vegetation.
B4-Red	665	30	10	Maximum chlorophyll absorption
B5-Vegetation Red Edge	705	15	20	Position of red edge; consolidation of atmospheric corrections / fluorescence baseline.
B6-Vegetation Red Edge	740	15	20	Position of red edge, atmospheric correction, retrieval of aerosol load
B7-Vegetation Red Edge	783	20	20	Leaf Area Index (LAI), edge of the NIR plateau
B8-NIR	842	115	10	LAI
B8A-Narrow NIR	865	20	20	NIR plateau, sensitive to total chlorophyll, biomass, LAI and protein; water vapour absorption reference; retrieval of aerosol load and type.
B9-Water vapour	945	20	60	Water vapour absorption, atmospheric correction.
B10-SWIR-Cirrus	1380	30	60	Detection of thin cirrus for atmospheric correction.
B11-SWIR	1610	90	20	Sensitive to lignin, starch and forest above ground biomass. Snow/ice/cloud separation.
B12-SWIR	2190	180	20	Assessment of Mediterranean vegetation conditions. Distinction of clay soils for the monitoring of soil erosion. Distinction between live biomass, dead biomass and soil, e.g. for burn scars mapping.

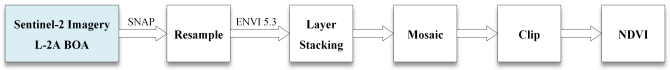

Fig. 3. Processing flowchart of Sentinel-2 imagery.

were collected and divided into four sample sets: roots, stems, leaves, and grains. The samples were cured at 105 °C for 60 min and then dried at a constant temperature of 85 °C for 48 h to obtain the dry weights of different sample sets. The crop yield determination method involved counting the number of wheat plants within one meter along a ridge and measuring the distance between two ridges. Then, several wheat plants near the sampling point were collected, naturally dried and weighed, and the yield of each sampling point was calculated.

2) Acquisition and Processing of Time Series of Sentinel Images: Considering the availability of images and spatial resolution of data in the study area, this article selected the Level-2A products of Sentinel-2 A/B imagery as the main remotely sensed data sources (data acquisition website: https://scihub.copernicus.eu/); these data are released by the European Space Agency (ESA) and can provide orthorectified bottom-of-atmosphere (BOA) reflectance with subpixel multispectral registration. Sentinel-2 carries a multispectral instrument (MSI), which is divided into two satellites (2A and 2B), at a height of 786 km, covering 13 spectral bands and a width of 290 km. The ground resolution is 10, 20, and 60 m [51], the revisit frequency of each satellite is 10 days and the combined constellation revisit time is 5 days. To meet the temporal resolution requirements for crop yield estimation, in this study, 12 phases of Sentinel-2 imageries in 2019 were selected; the images cover all growth and development stages of wheat in the study area. The spectral band characteristics of sentinel-2 satellite imagery and the series time of the Sentinel-2 data are shown in Tables II and III. The processing of Sentinel-2 data is shown in Fig. 3., and the normalized differential vegetation index (NDVI) was used to retrieve the

TABLE III
TIME SERIES OF SENTINEL-2 DATA FROM THE STUDY AREA

Imagery Number	Imagery Date
N1	2019-04-15
N2	2019-04-25
N3	2019-05-03
N4	2019-05-10
N5	2019-05-28
N6	2019-06-04
N7	2019-06-17
N8	2019-06-19
N9	2019-06-22
N10	2019-07-27
N11	2019-08-03
N12	2019-08-31

LAI through a statistical model. The NDVI and LAI can be expressed as

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_R}{\rho_{\mathrm{NIR}} + \rho_R} \tag{1}$$

$$\mathrm{LAI} = 10.575\mathrm{NDVI} - 6.0645 \tag{2}$$

where gρ_{NIR} and ρ_R are the near-infrared and red wavelength reflectance, respectively. The results of LAI retrieval are shown in Fig. 4.

3) Meteorological Data and Other Spatial Data: Meteorological data are important drivers for crop yield estimation models. The meteorological data were collected at the China Meteorological Data Service Center (http://data.cma.cn/en), mainly include the average temperature (°), maximum temperature (°), minimum temperature (°), precipitation (mm), relative humidity (%), average wind speed (m/s), average sea level pressure (kPa), and solar duration (h). The temporal resolution of the obtained meteorological data is one day.

This article performs crop yield estimation for wheat plots. Other spatial data mainly include field survey data and basic data. The field survey was conducted to measure some model parameters and yield data to verify the accuracy of the coupled

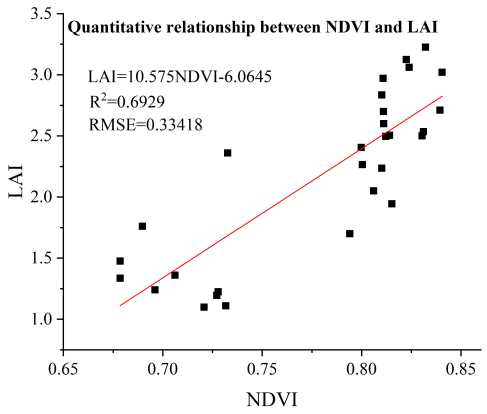

Fig. 4. Relationship between NDVI and LAI of spring wheat in study area.

model simulation results. Additionally, the basic data used in this article mainly include the study area plot boundary vector data and crop distribution maps. The parcel boundary vectors were drawn using RapidEye data with a spatial resolution of 6.5 m, and the crop map was obtained based on the farming plan and the classification results of RS images.

C. Models

1) CASA Model: The CASA model [53] is one of the most widely used LUE models, and the corresponding yield estimation model is based on a combination of biomass and HI information

$$\text{yield} = \frac{\text{Biomass} \times \text{HI}}{1 - \theta_{\text{grain}}} \tag{3}$$

where Biomass is the gross biomass during the crop growing season, HI is the harvest index, and θ_{grain} is the moisture content of the crop grains. The value of the HI in the study area was 0.47 [54], and the moisture content of wheat seeds in the study area was 12.5% [55]. The conversion between biomass and yield is achieved through the HI, and the biomass during the crop growing season is expressed by the following equation:

$$\text{Biomass} = \sum_{t=t_0}^{t_n} \text{NPP}(x,t) = \sum_{t=t_0}^{t_n} (\text{APAR}(x,t) \times \varepsilon(x,t)) \tag{4}$$

where Biomass is the gross biomass during the crop growing season, NPP is the NPP during the crop growing season, t is the different growth periods of the crop with a unit of one day, x is the spatial position of the crop growth, t_0 is the beginning of the crop growth stage, t_n is the crop harvesting stage, APAR(x,t) is the absorbed photosynthetic active radiation in MJ/m^2, and $\varepsilon(x,t)$ is the actual LUE in gC/MJ. APAR(x,t) and $\varepsilon(x,t)$ are expressed by the following equations:

$$\text{APAR}(x,t) = \text{SOL}(x,t) \times \text{FPAR}(x,t) \times 0.5 \tag{5}$$

$$\varepsilon(x,t) = T_{\varepsilon 1}(x,t) \times T_{\varepsilon 2}(x,t) \times W_{\varepsilon}(x,t) \times \varepsilon_{\max} \tag{6}$$

where SOL(x, t) is the total solar radiation (SOL) at the surface in MJ/m^2, 0.5 is the ratio of solar radiation available to vegetation, and FPAR(x, t) is the fraction of photosynthetically active radiation. $\varepsilon(x,t)$ is the actual LUE in gC/MJ. $T_{\varepsilon 1}(x,t)$ and $T_{\varepsilon 2}(x,t)$ are temperature stress factors, which indicate the degrees of low-temperature and high-temperature stress on the crop LUE. $W_{\varepsilon}(x,t)$ is the water stress factor, and $\varepsilon_{\max}$ is the maximum LUE of the crop without any restrictions.

2) CASA Model Parameter Calibration: Fraction of Photosynthetic Active Radiation (FPAR): Los, Field, *et al.* [56], [57] compared simulated FPAR values with measured FPAR values and found that the FPAR values simulated based on the NDVI were higher than the actual FPAR values; additionally, the FPAR values simulated based on the ratio vegetation index (SR) were lower than the actual FPAR values. Los *et al.* [57] introduced a correction coefficient α with a value range of 0.1–0.9. In the test results, when $\alpha = 0.5$, the measured value yields the best correlation with FPAR. Therefore, FPAR can be expressed by the following equation:

$$\text{FPAR}_{\text{NDVI}}(x,t) = \frac{(\text{NDVI}(x,t) - \text{NDVI}_{\min})(\text{FPAR}_{\max} - \text{FPAR}_{\min})}{(\text{NDVI}_{\max} - \text{NDVI}_{\min})} + \text{FPAR}_{\min} \tag{7}$$

$$\text{FPAR}_{\text{SR}}(x,t) = \frac{(\text{SR}(x,t) - \text{SR}_{\min})(\text{FPAR}_{\max} - \text{FPAR}_{\min})}{(\text{SR}_{\max} - \text{SR}_{\min})} + \text{FPAR}_{\min} \tag{8}$$

$$\text{SR}(x,t) = \frac{1 + \text{NDVI}(x,t)}{1 - \text{NDVI}(x,t)} \tag{9}$$

$$\text{FPAR}(x,t) = \frac{\text{FPAR}_{\text{NDVI}}(x,t) + \text{FPAR}_{\text{SR}}(x,t)}{2} \tag{10}$$

where FPAR$_{\text{NDVI}}(x,t)$ is the FPAR result estimated based on the NDVI and FPAR$_{\text{SR}}(x,t)$ is the FPAR result estimated based on the SR. In addition, the values of NDVImax and NDVImin in this article are 95% and 5% of the lower quantile of the NDVI, respectively. Based on the parameters proposed by Perman *et al.* [58], the SOL was estimated using parameters such as the latitude, solar duration, and day sequence.

Calculation of actual light use efficiency (ε): LUE (ε) refers to the efficiency of vegetation in converting absorbed photosynthetically active radiation (APAR) to organic carbon through canopy photosynthesis, which reflects the ability of vegetation to convert solar energy into organic matter. Potter *et al.* [53] noted that vegetation can achieve a maximum LUE under ideal conditions, but with changes in environmental conditions during the growing season, the LUE of vegetation also changes and is mainly affected by temperature and moisture factors. In this study, the optimum temperature for wheat growth was set at 20 °C. It is well known that wheat is a C_3 crop [59], and its maximum LUE can reach an upper limit of 3.0 gC/MJ. In this article, the maximum LUE is set to 2.6 gC/MJ based on the characteristics of the climate and vegetation types in the study area [60].

3) WOFOST Model: The WOFOST model is a crop growth model that was jointly developed by Wageningen University and Research Centers [61]. It is a process-oriented, mechanism-based model that simulates the growth process of

TABLE IV
VALUES OF INPUT CROP PARAMETERS FOR THE WOFOST MODEL

Parameters	Description	Values	Unit	Acquired method
TSUM1	Thermal time from emergence to anthesis	672	℃•d^{-1}	Meteorological data
TSUM2	Thermal time from anthesis to maturity	891	℃•d^{-1}	Meteorological data
TDWI	Initial total crop dry weight	130	kg/ha	Field observation
TBASEM	Lower threshold temperature for emergence	5	℃	Meteorological data
TEFFMX	Maximum threshold temperature for emergence	30	℃	Meteorological data
Q10	Relative change in respiration rate per 10 ℃ temperature change.	2	—	WOFOST default
FO1(DVS=0.1)	Fraction of ground dry matter to storage organ (DVS=0.1)	0	kg/kg	Field observation
FO2(DVS=0.5)	Fraction of ground dry matter to storage organ (DVS=0.5)	0	kg/kg	Field observation
FO3(DVS=1.1)	Fraction of ground dry matter to storage organ (DVS=1.1)	0.23	kg/kg	Field observation
FO4(DVS=1.7)	Fraction of ground dry matter to storage organ (DVS=1.7)	0.76	kg/kg	Field observation
CVL	Conversion efficiency of assimilates into leaf	0.68	kg/kg	Field observation
CVS	Conversion efficiency of assimilates into stem	0.64	kg/kg	Field observation
CVO	Conversion efficiency of assimilates into storage organ	0.82	kg/kg	Field observation
CVR	Conversion efficiency of assimilates into root	0.69	kg/kg	Field observation
RMR	Relative maintenance respiration rate roots	0.01	mm	WOFOST default
RMO	Relative maintenance respiration rate storage organs	0.009		WOFOST default
RML	Relative maintenance respiration rate leaves	0.025	—	WOFOST default
RMS	Relative maintenance respiration rate stems	0.015	day	WOFOST default
JUL	Sowing date	25, April-10,May	day	Statistical data
AMAX	Maximum leaf CO_2 assimilation rate	25-75	kg(CO_2)/ ha(leaf)/h	WOFOST default

TABLE V
CALIBRATION OF THE MAIN SOIL PARAMETERS FOR THE WOFOST MODEL

Parameters	Description	Values	Unit	Acquired method
NBASE	Mean basic soil nitrogen content	249	mg/kg	Field observation
PBASE	Mean basic phosphorus content	33	mg/kg	Field observation
KBASE	Mean basic potassium content	240	mg/kg	Field observation
NF	Quantity of nitrogen fertilizer	53	kg/ha	Field observation
PF	Quantity of phosphorus fertilizer	83	kg/ha	Field observation
KF	Quantity of potassium fertilizer	27	kg/ha	Field observation
SMFCF	Soil moisture content at field capacity	0.33	cm^3/cm^3	Statistical data
SMW	Soil moisture content at wilting point	0.05	cm^3/cm^3	Statistical data
SM0	Soil moisture content of saturated soil	0.44	cm^3/cm^3	Statistical data

crops with a one-day step and integrates crop, soil, climate, and management information, as well as data for other factors. The WOFOST model mainly includes three parts: a crop growth simulation module, a soil moisture simulation module, and a crop evapotranspiration module. These modules encompass the entire growth simulation process of crop respiration, photosynthesis, growth, biomass accumulation and distribution in the WOFOST model. Based on CO_2 as the core driving factor, the model first calculates the potential yield through the crop growth simulation module and the crop evapotranspiration module; then, the attainable yield is determined based on the water and nutrient restrictions, and crop growth respiration and transpiration are removed to obtain the net assimilation amount. Finally, the final crop yield is calculated based on the storage organ partitioning coefficient.

The WOFOST model has been applied in China for more than 10 years, and the main parameters of the model have been fully calibrated under different meteorological, soil, and management conditions through ground observations and RS data assimilation [62]. However, many factors can influence the parameters of crop models, and even the crop parameters for different varieties of the same crop can display relatively large differences. Therefore, it is necessary to calibrate the core parameters of crop models.

4) WOFOST Model Calibration: The WOFOST crop model has achieved relatively satisfactory results in Northeast China [62], [63]. However, for different research areas, due to differences in the ecological environment, climatic conditions, and different soil and management conditions, parameters need to be calibrated; otherwise, the applicability of the model will be reduced. In this article, the parameters of the WOFOST model are calibrated through a combination of a literature survey [64], remotely sensed data and field measurements. The calibration results are shown in Tables IV and V.

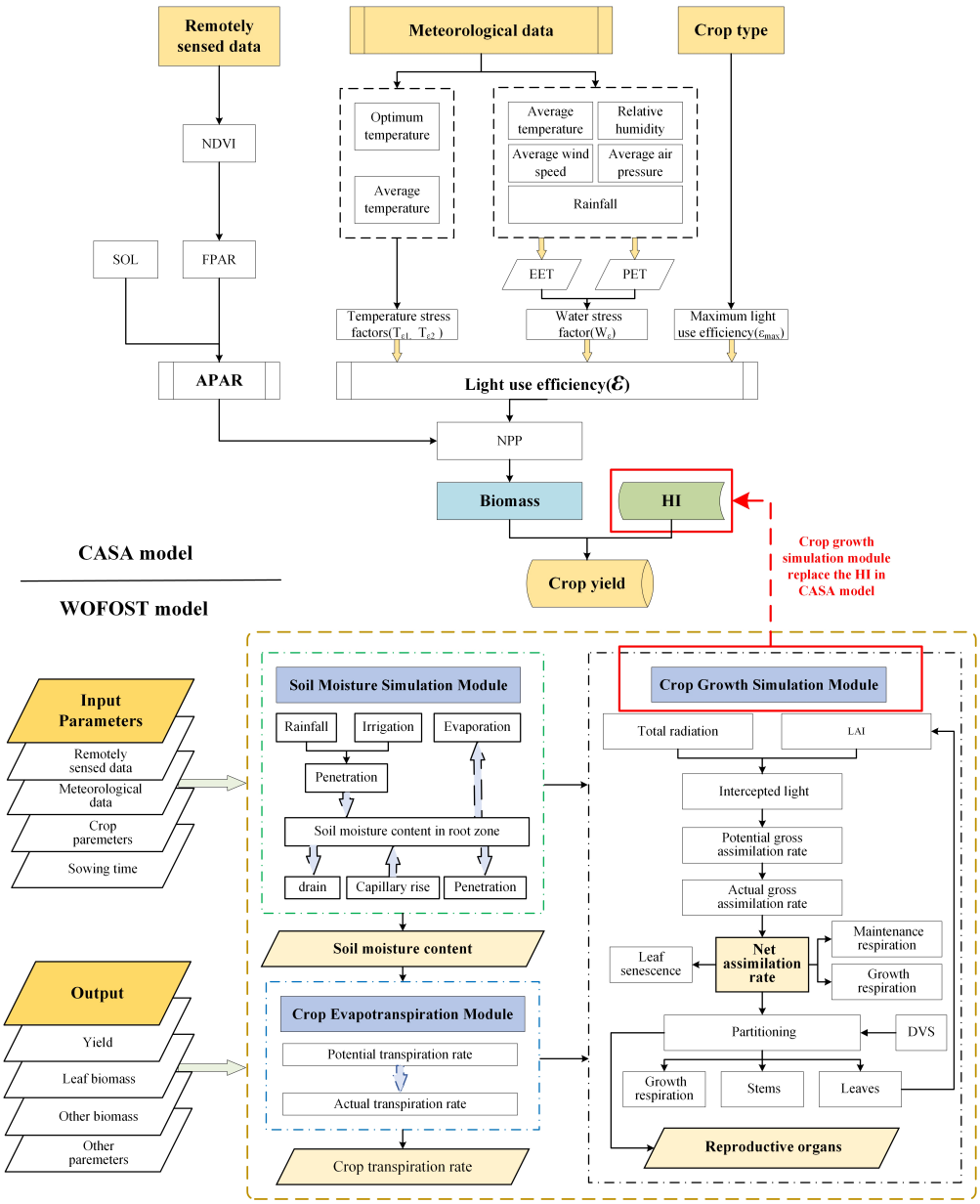

Fig. 5. Flowchart of the CASA-WOFOST coupled model for estimating the regional crop yield.

D. Data Assimilation Algorithm

Crop models can simulate the growth and development of crops at a point-based scale, and remotely sensed data assimilation technology can extrapolate crop models from the point-based scale to a regional scale. The parameters retrieved from RS can be integrated into the crop model simulation process to achieve regional-scale crop growth, which can solve the problem that the initial parameters of the crop model at the regional scale are difficult to accurately obtain [65]. In this study, we use the ensemble Kalman filter (EnKF) [66] method to assimilate the LAI into the crop model. The EnKF method is based on the concept of ensemble forecasting and uses a state ensemble to represent the probability density function, which is essentially a Monte Carlo approximation of the Kalman filter. Additionally, EnKF overcomes the problem of processing nonlinear data with a Kalman filter while reducing the number of calculations compared to that in the variational method. The assimilation process of the EnKF algorithm is divided into two steps: forecasting and updating. The EnKF algorithm is expressed by the following equation:

$$A_a = A_f + K \times (D_t - \mathrm{HA}) \tag{11}$$

$$K = A_c H^T \times \left(\mathrm{HA}_c H^T + D_c\right)^{-1}, \quad k \in R^{n \times N} \tag{12}$$

where A_a is the optimal estimated ensemble, A_f is an ensemble of the forecasts, K is the Kalman gain matrix, D_t is the ensemble of the observations, H is the nonlinear operator, A is obtained by the prediction equation, which denotes the crop model producing predictions of the same parameters present in the observations, and HA is typically equal to 1.

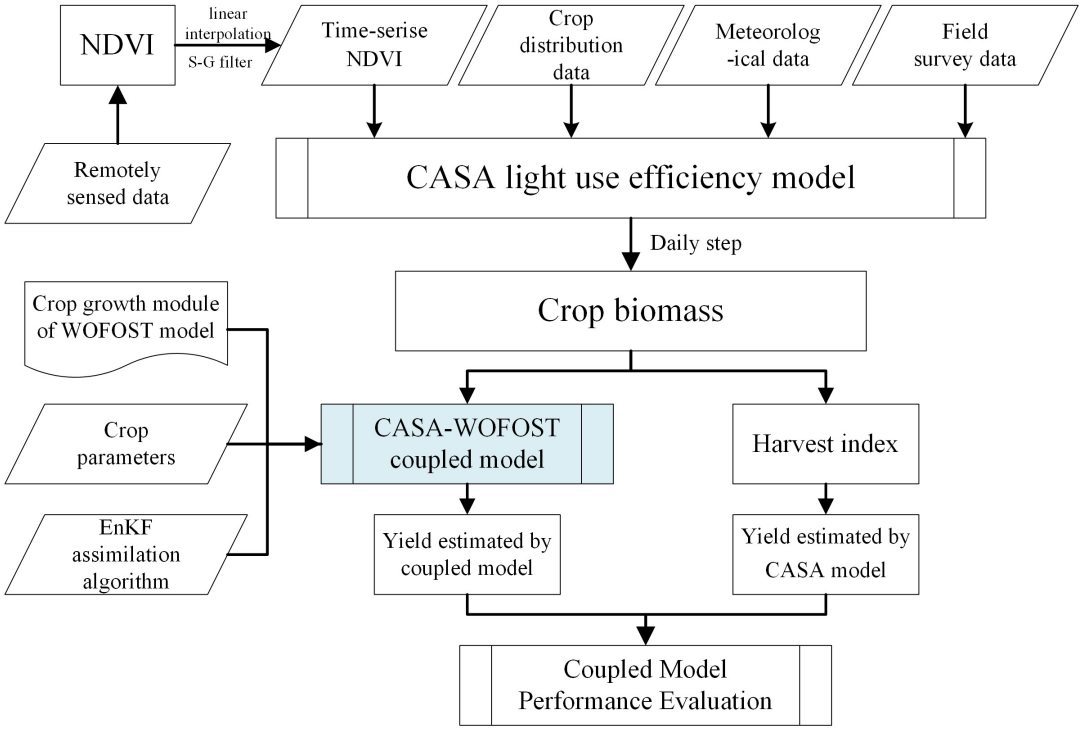

Fig. 6. Flowchart of the methods used in this study.

E. CASA-WOFOST Coupled Model

Since the CASA model was first established, it has been widely used to estimate the NPP of terrestrial ecosystems at the global scale. The application of the CASA model to the monitoring of crops at small regional scales and even field scales is relatively rare. The CASA model is relatively simple and practical, and few parameters must be obtained and calibrated. However, the CASA model ultimately simulates the NPP of a crop. We need to construct a HI to achieve the conversion between NPP and the crop yield, but from photosynthetic assimilation to biomass accumulation and the final crop yield, crop growth involves many complex physiological and biochemical processes that cannot be fully reflected by a simple HI. Although the WOFOST model can simulate crop growth in detail, many model parameters are required. In this article, after simulating the biomass of crops with the CASA model, we use the crop growth simulation module in the WOFOST model instead of the HI to simulate the crop yield. The flowchart of the CASA-WOFOST coupled model for estimating crop yields is shown in Fig. 5. In this study, we first use the linear interpolation method [67] and S-G (Savitzky-Golay) iterative filtering method [68] to obtain NDVI time series for crops based on remotely sensed data. Then, by using crop distribution data, meteorological data, and so on, we run the CASA model with a daily time step to simulate daily crop biomass. Finally, by coupling the crop growth module of the WOFOST model with the CASA model, the wheat yield in the study area was simulated and then compared with the yields simulated separately by the CASA model and WOFOST model. The modeling flowchart is shown in Fig. 6.

F. Coupled Model Simulation Accuracy Evaluation Index

The simulation accuracy of the coupled model proposed in this study is assessed based on the root mean square error (RMSE), normalized root mean square error (NRMSE), Nash-Sutcliffe efficiency coefficient (NSE), and coefficient of determination (R^2) based on linear regression. The RMSE, NRMSE, and NSE are given by

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X - X_{\mathrm{obs}})^2} \tag{13}$$

$$\mathrm{NRMSE} = \begin{cases} \mathrm{NRMSE}_1 = \dfrac{\mathrm{RMSE}}{\overline{X_{\mathrm{obs}}}} \\ \mathrm{NRMSE}_2 = \dfrac{\mathrm{RMSE}}{X_{\mathrm{obs\,max}} - X_{\mathrm{obs\,min}}} \\ \mathrm{NRMSE}_3 = \dfrac{\mathrm{RMSE}}{\sigma} \\ \mathrm{NRMSE}_4 = \dfrac{\mathrm{RMSE}}{Q_1 - Q_3} \end{cases} \tag{14}$$

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{n}(X - X_{\mathrm{obs}})^2}{\sum_{i=1}^{n}\left(X_{\mathrm{obs}} - \overline{X}\right)^2} \tag{15}$$

where n is the number of observations, X is the simulated value of the coupled model, X_{obs} is an observation, σ is the standard deviation of observations, Q_1 and Q_3 are the 25th and 75th percentile of observations. The NRMSE is a function that allows the user to calculate the absolute value between predictions and observations; there are four different types of normalization methods. The lower the RMSE is, the higher the simulation accuracy; the lower the NRMSE

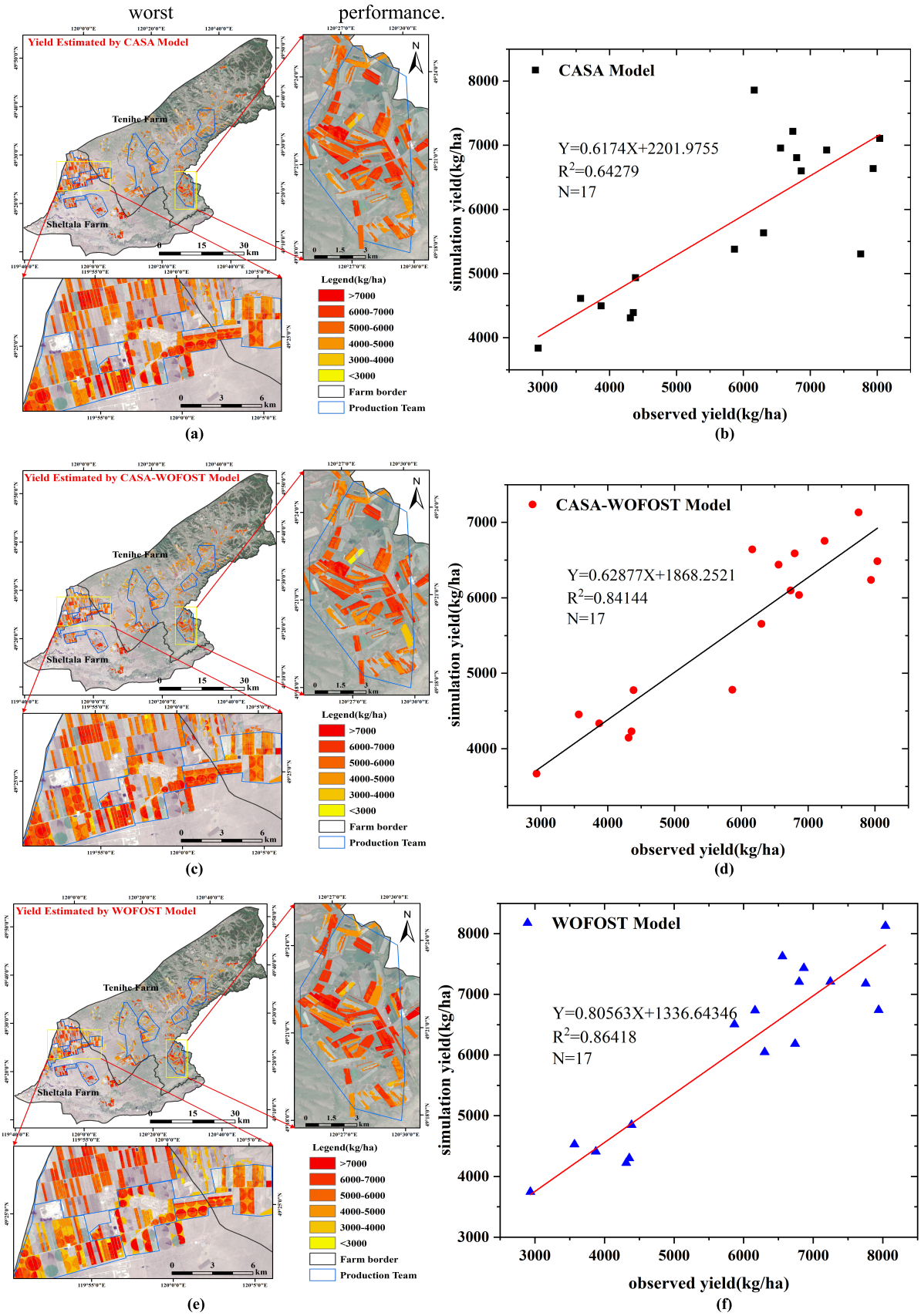

Fig. 7. Spatial distribution map of yield simulations by different models and the corresponding coefficient of determination values. (a) Spatial distribution map of the yield simulated by the CASA model. (b) Coefficient of determination of the CASA model. (c) Spatial distribution map of the yield simulated by the CASA-WOFOST coupled model. (d) Coefficient of determination of the CASA-WOFOST coupled model. (e) Spatial distribution map of the yield simulated by the WOFOST model. and (f) Coefficient of determination of the WOFOST model. The background is the Sentinel-2A imagery acquired on June 19, 2019, shown in a quasi-natural color band composition of R(4) G(3) B(2) with standard deviation stretching.

TABLE VI
ANALYSIS RESULTS OF WHEAT YIELD SIMULATION IN SHELTALA-TENIHE FARM IN 2019

Accuracy evaluation index	CASA model	CASA-WOFOST model	WOFOST model
RMSE	948.32 kg/ha	792.11 kg/ha	623.64 kg/ha
$NRMSE_1$	0.162	0.135	0.106
$NRMSE_2$	0.186	0.155	0.122
$NRMSE_3$	0.599	0.5	0.394
$NRMSE_4$	0.372	0.310	0.244
NSE	0.641	0.759	0.847
R^2	0.64	0.84	0.86
Absolute error	715.53 kg/ha	655.93 kg/ha	522.59 kg/ha
Relative error	12.77%	11.54%	9.97%

is, the higher the simulation accuracy; the closer the NSE is to 1, the higher the simulation accuracy.

III. RESULTS

A. Yield Simulation Accuracy Evaluation

Fig. 7 shows the yield simulation results, and the spatial distribution maps of different models are shown in Fig. 7(a), (c), and (e). A total of 17 sampling quadrats were observed in the study area for the model accuracy evaluation. Then, we used the R^2, RMSE, NRMSE, NSE, absolute error, and relative error results to evaluate the performance of the CASA model, CASA-WOFOST coupled model, and WOFOST model in yield simulation. This study mainly used the CASA-WOFOST coupled yield estimation model to simulate the yield of wheat at the Sheltala and Tenihe Farms, and the results were compared with those for the individual CASA and WOFOST models. The coefficient of determination results is shown in Fig. 7. (b), (d), and (f). The R^2 values of the CASA model, coupled model, and WOFOST model are 0.64, 0.84, and 0.86, respectively. Additionally, an analysis of different accuracy evaluation indicators is shown in Table VI. The results show that the RMSEs of the CASA model, CASA-WOFOST coupled model, and WOFOST model are 948.32, 792.11, and 623.64 kg/ha, respectively; the $NRMSE_1$s are 0.162, 0.135, and 0.106, respectively; the $NRMSE_2$s are 0.186, 0.155, and 0.122, respectively; the $NRMSE_3$s are 0.599, 0.5, and 0.394, respectively; the $NRMSE_4$s are 0.372, 0.31, and 0.244, respectively; the NSEs are 0.641, 0.759, and 0.847, respectively; the average absolute errors are 715.53, 655.93, and 522.59 kg/ha, respectively; and the average relative errors are 12.77%, 11.54%, and 9.97%, respectively. All of the above results show that the WOFOST model performs best in yield estimation, followed by the CASA-WOFOST coupled model, and the CASA model yields the worst performance.

B. Analysis of the Yield Simulated by Different Models

An analysis of the yield distribution maps [Fig. 7(a), (c), and (e)] indicates that the yields of Sheltala Farm are higher than those of Tenihe Farm, and the yields in the south are higher than those in the north, which is supported by the observed yields. In addition, through the accuracy evaluation, we selected three representative fields in the study area, and the spatial distribution and digital elevation model (DEM) ranges of the three fields can be seen in Fig. 8. The accuracy of the sampling quadrats for these three representative fields is shown in Table VII. For the Field1 sampling quadrats, whether the CASA model, WOFOST model, or CASA-WOFOST coupled model is applied, when the value of the DEM is high or a point is close to a mountainous area, the simulation accuracy of the model is shown below. However, at these points, the CASA-WOFOST coupled model and WOFOST model usually produce higher simulation accuracies than the CASA model. In contrast, when the terrain is relatively flat, the elevation is low, or the elevation change is minimal, the accuracies of the CASA model and the CASA-WOFOST coupled model are similar, as demonstrated by the accuracy results for Field2 and Field3 in Table VII.

C. Comparative Analysis of the Model Operating Efficiency

As mentioned earlier, we used Sentinel-2 imageries with a 10-m spatial resolution in daily crop growth simulations; approximately 74 million pixel calculations were performed each day for the study area, and the growing season of wheat was approximately 120 days; therefore, more than 8.8 billion pixel operations were executed during the entire growing season. Therefore, to compare the operating efficiencies of the CASA model, CASA-WOFOST coupled model, and WOFOST model, we implemented these three models on computers with the same configuration environment, which was an Intel Core i7-8700 3.2 GHz, 6-core CPU with 32 GB RAM. The model run time is shown in Table VIII, and the operating times of the CASA model, CASA-WOFOST coupled model, and WOFOST model over the growing season of spring wheat are 37 min, 48 min, and 1 day 5 h 7 min, respectively. The results suggest that the CASA-WOFOST coupled model can greatly improve the operating efficiency compared with that of the WOFOST model; thus, the coupled model can be applied in cases with large farms and high spatial resolution images while ensuring sufficient simulation accuracy.

IV. DISCUSSION

The CASA model was used to estimate the NPP of terrestrial ecosystems at large regional or global scales in the early years of establishment [16], [69], [70]. In previous studies, the CASA model was seldom applied at the field scale to estimate the yield, so we first used the CASA model to estimate the crop yield at the field scale; however, the simulation accuracy was not very high. However, the crop growth

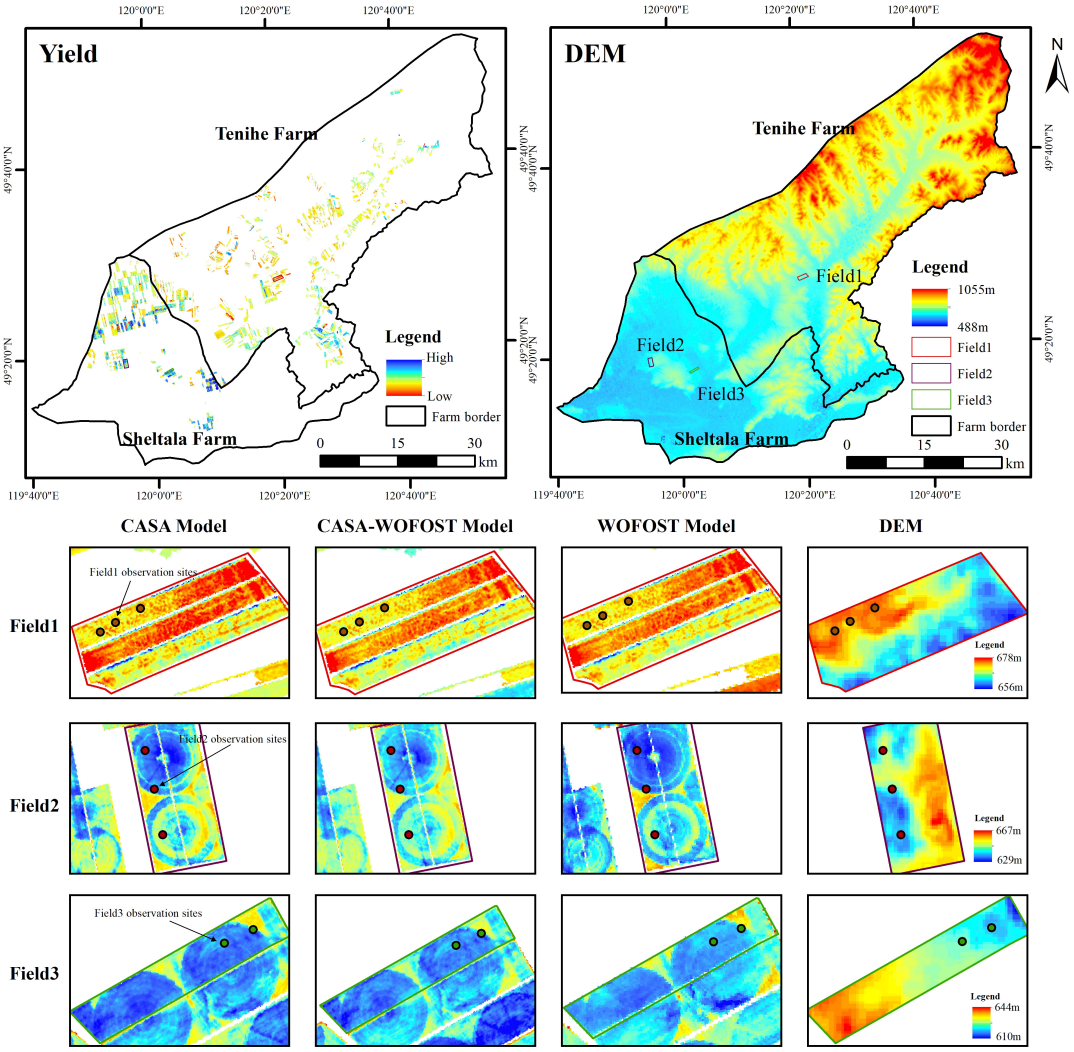

Fig. 8. Representative fields with sampling quadrats were selected to analyze the relationship between the yield distribution and DEM. The upper-left map shows the yield distribution in the study area, the upper-right map shows the DEM in the study area. In the next rows, the distribution maps of the three fields is shown, and from left to right are the simulated yield results of the CASA model, CASA-WOFOST coupled model, and WOFOST model followed by the DEM.

TABLE VII

ACCURACY ANALYSIS OF THE YIELD IN THE SAMPLING QUADRAT FIELDS

Field name	OY[a]	CM[b]	CWM[c]	WM[d]	CM		CWM		WM		DEM
					AE[e]	RE[f]	AE	RE	AE	RE	
Field 1	3567.85	4612.91	4453.08	4526.97	1045.06	29.29%	885.23	24.81%	959.12	26.88%	674
	4387.25	4934.07	4776.78	4848.99	546.82	12.46%	389.53	8.88%	461.74	10.52%	673
	3873.88	4496.08	4335.32	4409.71	622.21	16.06%	461.44	11.91%	535.83	13.83%	676
Field 2	8037.79	7105.31	6483.21	8126.98	932.48	11.60%	1554.58	19.34%	89.19	1.11%	650
	6556.95	6954.10	6437.99	7625.69	397.15	6.06%	118.96	1.81%	1068.74	16.30%	642
	6301.64	5633.85	5654.77	6047.18	667.79	10.60%	646.87	10.27%	254.46	4.04%	637
Field3	7246.08	6923.51	6753.94	7208.51	322.57	4.45%	492.14	6.79%	37.57	0.52%	620
	6797.71	6804.57	6587.06	7205.03	6.86	0.10%	210.65	3.10%	407.32	5.99%	625

[a] "OY" denotes the observed yield (unit: kg/ha); [b] "CM", [c] "CWM" and [d] "WM" denote the CASA model-simulated yield, CASA-WOFOST coupled model-simulated yield, and WOFOST model-simulated yield, respectively (unit: kg/ha); [e] "AE" and [f] "RE" denote the absolute error and relative error (unit of "AE" is kg/ha), respectively.

model can simulate the entire process of crop growth and the corresponding relationship with the surrounding environment from the perspective of ecological physiology considering crop, soil, climate, and management information, as well as other factors, to fully simulate physiological processes such as crop respiration, photosynthesis, phenological actions, biomass accumulation, and distribution, but there are so many input parameters in crop growth models that the growth and development of crops can only be simulated at the point scale. If we want to extrapolate a crop model from a point-based

TABLE VIII
COMPARISON OF THE OPERATING TIMES OF DIFFERENT MODELS

Model	Model operating time
CASA model	37 minutes
CASA-WOFOST coupled model	48 minutes
WOFOST model	1 day 5 hours 7 minutes

scale to a regional scale, we need to use remotely sensed data assimilation technology, which involves extensive calculations [48], [71].

In this article, based on comparing and analyzing the advantages and limitations of the current mainstream yield estimation models, we proposed an accurate and efficient CASA-WOFOST coupled crop yield estimation model. First, we modified the CASA model by using two VIs, the NDVI and SR, to calculate fPAR instead of establishing a linear relationship between fPAR and SR in the original CASA model. Additionally, we successfully generated the daily NDVI through the NDVI time series reconstruction algorithm, which has the same time step as the WOFOST model to effectively couple the CASA and crop growth models. Second, we used the modified CASA model as the core of the coupled model and introduced the dry matter distribution process in the crop growth simulation module of the WOFOST model to replace the HI in the CASA model, thus avoiding the unnecessary errors associated with the HI and improving the accuracy of the yield estimation model. The new CASA-WOFOST yield estimation model can not only take advantage of the benefits of the CASA model and provide a fast operating speed but also take advantage of the mechanistic advantages of the WOFOST model.

Based on the new CASA-WOFOST coupled yield estimation model, we successfully simulated the wheat yield from time series of Sentinel-2 remotely sensed imageries of the Sheltala and Tenihe Farms in the Inner Mongolia Autonomous Region in 2019. To evaluate the performance of the coupled model in yield simulation, we used the individual CASA and WOFOST models to simulate the yield in the study area under the same conditions. The comparison of the estimated results of the three models with field campaign data indicated that the CASA-WOFOST coupled model performs better than the CASA model based on all evaluation indexes. Although the accuracy of the coupled model was not as high as that of the WOFOST model, the difference between the two models was minimal; additionally, the coupled model displayed a much faster running speed than the WOFOST model. Moreover, the coupled model has the highest simulation accuracy in mountainous areas and regions of uneven terrain.

The CASA-WOFOST coupled yield estimation model applied in this article can improve the accuracy and efficiency of crop yield estimation and provide a reference for regional crop yield estimation, but there are also limitations. First, in this study, as the key parameter of the assimilation algorithm, LAI was used to update the parameter in the crop model, so the accuracy of LAI retrieval directly affects the accuracy of yield simulation. Considering that the natural environment and farming methods inside the experimental fields are relatively stable, and the focus of this study was on the accuracy of yield simulation, we used empirical statistical models for LAI retrieval and applied it for data assimilation and yield simulation. The reason for choosing the statistical model was that it could provide a higher LAI retrieval accuracy than the mechanism model for the selected study area. However, because the empirical statistical model is designed for a specific environment, the model needs to be rebuilt when the application time, area, and crop type change [72], [73], which limits the general application of the coupled model; the inclusion of the mechanism model can overcome this problem to a certain extent. Second, the CASA-WOFOST coupled model has only been applied in small areas so far, and the simulation accuracy of yields over large areas remains uncertain. The large-scale application of the coupled model should be considered in the next study.

V. CONCLUSION

In this article, the CASA-WOFOST coupled model was used to simulate the wheat yield while comprehensively considering environmental factors such as the LUE of crops and the entire process of crop growth and development. The results showed that the R^2 of the coupled model is 0.84, the RMSE is 792.11 kg/ha, the NSE coefficient is 0.759, the average absolute and relative errors are 655.93 kg/ha and 11.54%, respectively, and the four types of NRMSE are 0.135, 0.155, 0.5, and 0.31, respectively. In addition, the CASA-WOFOST coupled model displayed a higher simulation accuracy in mountainous areas and regions of uneven terrain compared to other models and greatly improved the operating efficiency. Overall, the yield simulation results of the coupled model were better than those based on the CASA model and HI. The CASA-WOFOST coupled model used at the regional field scale yields high simulation accuracy and can provide theoretical support for crop yield estimation under the local conditions.

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